Construction Chemicals

Product Datasheet
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Protective Coating

O-Guard FR

Water-Based Intumescent
Fire-Protective Coating for Steel



Product Description

O-Guard FR is a water-based acrylic intumescent coating for the fire protection of interior structural steel. When exposed to fire it expands to form an insulating char that delays the temperature rise of the steel, providing fire ratings of up to 120 minutes depending on the applied thickness and the steel section.

Uses

- Interior structural steel beams and columns.
- Warehouses and industrial buildings.
- Shopping malls, offices and public buildings.
- Steel structures requiring 60, 90 or 120 minutes fire rating.

Advantages

- Fire protection up to 120 minutes.
- Water-based, low odour.
- Thin-film, decorative white finish.
- Applied by brush, roller or airless spray.
- Can receive a decorative top coat.

Application

1. **Surface Preparation**
   Blast-clean steel and apply an approved anti-corrosive primer; clean and dry before coating.
2. **Mixing**
   Stir thoroughly with a slow-speed mixer; use as supplied.
3. **Application**
   Apply in several coats to the dry film thickness required for the fire rating; apply an optional top coat.

Consumption

1.2 – 1.6 m²/kg
for 60 minutes fire rating
Thickness depends on steel section factor and required rating.

Packaging

5 kg and 20 kg pails.

Storage & Shelf Life

12 months in original, unopened packaging.
Store at +5 °C to +35 °C; protect from frost.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Base	Water-based acrylic intumescent	—
Finish / colour	Matt white	—
Density	1.3 – 1.4 kg/L	—
Solids content	65 – 75 %	—
Touch dry	1 – 2 hours	—
Recoat interval	6 – 8 hours	—
Fire rating	up to 120 minutes (interior steel)	—
Shelf life	12 months	—

Consumption

Fire rating	Coverage
60 minutes	1.2 – 1.6 m²/kg
90 minutes	0.9 – 1.1 m²/kg
120 minutes	0.6 – 0.8 m²/kg

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Relative humidity: max. 80 %.
- Interior use only; protect from water and condensation.
- Primer must be compatible and fully cured.

Application Notes

- Required thickness per member must be taken from the fire-protection schedule for the steel section factor.
- Apply in several coats; do not exceed the maximum wet thickness per coat.
- Measure the dry film thickness after each coat.
- A compatible decorative top coat may be applied after full drying.

Health & Safety

- Wear gloves, goggles and a mask during spray application.
- Avoid contact with eyes and prolonged skin contact.
- Ensure ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.
- Inhalation: move to fresh air.
- Ingestion: rinse mouth and seek medical advice.
- Clean tools and spray equipment with water immediately after use.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.